

Inference for Power-Law Lag Geometry under Noisy Transport Estimates

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Abstract

We study structural inference for lagged transport discrepancies that admit a power-law geometry $D_j = \zeta j^H$. The observation problem is coupled: each estimated discrepancy $\hat{D}_j$ carries finite-sample noise, and the leading variance of $\log \hat{D}_j$ depends on the same scale law that is being inferred. In this heteroscedastic log-domain model, we prove a central limit theorem for the two-step feasible weighted least-squares estimator of H , show that the oracle residual statistic has an exact $\chi^2(L-2)$ null law, and derive a minimax separation rate $(nL)^{-1/2}$ for distinguishing the power-law family from scale-inconsistent alternatives. The resulting procedure is adaptive in H and retains the optimal rate without prior knowledge of the exponent. Simulations confirm exact size calibration, FWLS-oracle agreement, and power consistent with the boundary.

1 Introduction

Let $\{P_t\}_{t \geq 0}$ be a family of probability measures on $\mathbb{R}^d$ evolving over time. The object of interest is the lagged transport geometry encoded by the Wasserstein-2 discrepancy curve

$$D_j := W_2(P_{t-j}, P_t), \quad j = 1, 2, \dots, L.$$

This curve is typically observed through estimators $\hat{D}_j$ constructed from n i.i.d. samples from each P_{t-j} and P_t , rather than through the population values D_j themselves. Under standard conditions on the measures [Fournier and Guillin, 2015, Goldfeld et al., 2024, del Barrio et al., 2023], the plug-in Wasserstein estimator satisfies

$$\hat{D}_j = D_j + \varepsilon_j, \quad \varepsilon_j \sim \text{AN}\left(0, \frac{\sigma_0^2}{n}\right), \quad (1)$$

where $\sigma_0^2 > 0$ is a nuisance constant depending on the local geometry of the measures.

A low-dimensional structural hypothesis is that the discrepancy curve follows a *power law*:

$$D_j = \zeta j^H, \quad \zeta > 0, H \in (0, 1). \quad (2)$$

This is scale consistency: the lag curve is compressed to the two-parameter family (ζ, H) , with exponent H interpreted as a roughness index of the latent drift in locally

stationary settings [Dahlhaus, 1997, Vogt, 2012, Tinio et al., 2025].

The inferential problem is not standard. In log-space, $\text{Var}(\log \hat{D}_j) \approx \sigma_0^2/(nD_j^2) = \sigma_0^2/(n\zeta^2 j^{2H})$, so the observation noise depends on the same scale law that is being inferred. The family is low-dimensional, but the noise is parameter-coupled.

The main results establish a complete inference layer for this family: a feasible WLS limit theorem for H , exact oracle calibration of a residual adequacy statistic, a minimax lower bound for distinguishing the family from scale-inconsistent alternatives, and adaptivity in H .

Main results.

- (1) **Feasible inference for the exponent.** Theorem 3.2 gives a central limit theorem for the two-step feasible WLS estimator $\hat{H}$ of H in the heteroscedastic log-scale model. The pilot-estimation error does not inflate the asymptotic variance beyond the oracle rate.
- (2) **Residual adequacy test.** Theorem 4.1 shows that the weighted residual sum of squares Q satisfies $Q \sim \chi^2(L-2)$ under the power-law family with oracle weights, and that two-step FWLS converges to the same law asymptotically as $n \rightarrow \infty$ with L fixed.
- (3) **Minimax boundary for model distinguishability.** Theorem 5.1 gives the minimax separation rate $(nL)^{-1/2}$ for distinguishing the power-law family from alternatives with scale inconsistency level $\kappa \geq \kappa^*$.
- (4) **Adaptive structural inference.** Theorem 6.1 shows that the FWLS adequacy procedure achieves the minimax rate without prior knowledge of H .
- (5) **Simulation signatures.** The numerical study reports exact size calibration, FWLS-oracle agreement, and power curves consistent with the boundary $\kappa^* = (nL)^{-1/2}$ (Section 7).

2 Model and Assumptions

2.1 Observation model

We observe a triangular array $\{\hat{D}_j\}_{j=1}^L$ satisfying

$$\hat{D}_j = D_j + \varepsilon_j, \quad \varepsilon_j \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{\sigma_0^2}{n}\right), \quad (\text{M})$$

where $D_j = \zeta j^H$ for unknown parameters $(\zeta, H) \in (0, \infty) \times (0, 1)$, $\sigma_0 > 0$ is known (or consistently estimable), and $n \geq 1$ is the effective sample size per lag.

Remark 2.1. The Gaussianity in (M) is justified for large n by the central limit theorem for Wasserstein estimators; see Goldfeld et al. [2024] and del Barrio et al. [2023]. The constant-variance assumption $\text{Var}(\varepsilon_j) = \sigma_0^2/n$ is the leading-order approximation; the full theory allows σ_j^2/n with $\sigma_j \leq C D_j$ bounded away from zero and infinity relative to D_j —see Assumption 2.3.

2.2 Log-scale regression

Taking logarithms of the power-law model and applying the delta method:

$$Y_j := \log \hat{D}_j = \alpha + H \log j + U_j, \quad (3)$$

where $\alpha = \log \zeta$ and

$$U_j = \log \left(1 + \frac{\varepsilon_j}{D_j} \right) \approx \frac{\varepsilon_j}{D_j} \sim \mathcal{N}(0, v_j), \quad (4)$$

$$v_j := \frac{\sigma_0^2}{n D_j^2} = \frac{\sigma_0^2}{n \zeta^2 j^{2H}}. \quad (5)$$

The variance v_j depends on H and ζ , which are the parameters of interest. This is the parameter-coupled heteroscedasticity that distinguishes our problem from standard regression.

2.3 Hypotheses

Let $\kappa \geq 0$ measure the *scale inconsistency*: under the alternative, each observation has an additional log-normal multiplicative factor,

$$\log \hat{D}_j = \alpha + H \log j + \kappa \xi_j + U_j, \quad \xi_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \xi_j \perp U_j.$$

The hypotheses are:

$$\begin{aligned} H_0 : \kappa &= 0 \quad (\text{scale consistency}), \\ H_1 : \kappa &\geq \kappa^* \quad (\text{scale inconsistency at level } \kappa^*). \end{aligned}$$

2.4 Assumptions

Assumption 2.2 (Signal strength). $\zeta \geq \zeta_{\min} > 0$ and $H \in [\underline{H}, \overline{H}] \subset (0, 1)$.

Assumption 2.3 (Noise regularity). $\sigma_0 \in [\sigma_{\min}, \sigma_{\max}]$ with $0 < \sigma_{\min} \leq \sigma_{\max} < \infty$. The delta-method approximation error satisfies $|U_j - \varepsilon_j/D_j| = O_p(n^{-1})$ uniformly in j .

Assumption 2.4 (Lag schedule). $L \geq 4$ is fixed as $n \rightarrow \infty$. The lags $x_j := \log j$ for $j = 1, \dots, L$ satisfy $\sum_j (x_j - \bar{x})^2 \geq c_L > 0$.

3 CLT for the Feasible WLS Estimator

3.1 Oracle WLS

With known weights $w_j^* = 1/v_j = n D_j^2/\sigma_0^2$, the oracle WLS estimator of $\beta = (\alpha, H)^\top$ in (3) is

$$\hat{\beta}^* = (\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^* \mathbf{Y}, \quad (6)$$

where $\mathbf{X} = [\mathbf{1} \mid \mathbf{x}]$, $\mathbf{x} = (\log 1, \dots, \log L)^\top$, and $\mathbf{W}^* = \text{diag}(w_1^*, \dots, w_L^*)$.

Proposition 3.1 (Oracle WLS CLT). *Under ?? 2.2–2.4, as $n \rightarrow \infty$ with L fixed,*

$$n^{1/2} (\hat{H}^* - H) \xrightarrow{d} \mathcal{N}(0, \sigma_0^2 [(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}]_{22}),$$

where $[\cdot]_{22}$ denotes the $(2, 2)$ entry. Explicitly,

$$\text{Var}(\hat{H}^*) \approx \frac{\sigma_0^2}{n} \cdot \frac{S_{w^*}}{S_{w^*} S_{w^* x^2} - (S_{w^* x})^2}, \quad (7)$$

where $S_{w^*} = \sum_j w_j^*$, $S_{w^* x} = \sum_j w_j^* x_j$, $S_{w^* x^2} = \sum_j w_j^* x_j^2$.

Proof. Standard weighted least squares with $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 n^{-1} (\mathbf{W}^*)^{-1})$; the $(2, 2)$ entry of $(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}$ follows by 2×2 matrix inversion. $\square$

3.2 Two-step feasible WLS

Since D_j is unknown, the oracle weights are infeasible. We use the following two-step procedure.

Algorithm 1 Two-Step FWLS Estimator

- 1: **Input:** $(\hat{D}_j)_{j=1}^L$, n , σ_0 .
 - 2: **Step 1 (Pilot OLS).** Compute $(\hat{\alpha}_{\text{pil}}, \hat{H}_{\text{pil}})$ by unweighted OLS of $\log \hat{D}_j$ on $(1, \log j)$.
 - 3: **Step 1b (Pilot imputation).** Set $\hat{D}_j^{\text{pil}} = \exp(\hat{\alpha}_{\text{pil}} + \hat{H}_{\text{pil}} \log j)$.
 - 4: **Step 2 (FWLS).** Set $\hat{w}_j = n(\hat{D}_j^{\text{pil}})^2/\sigma_0^2$ and compute $(\hat{\alpha}, \hat{H})$ by WLS with weights $(\hat{w}_j)$.
 - 5: **Output:** $\hat{H}$, $\hat{\alpha}$, residuals $\hat{U}_j = \log \hat{D}_j - \hat{\alpha} - \hat{H} \log j$.
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Theorem 3.2 (FWLS CLT). *Under ?? 2.2–2.4, the two-step FWLS estimator $\hat{H}$ satisfies, as $n \rightarrow \infty$ with L fixed:*

$$n^{1/2} (\hat{H} - H) \xrightarrow{d} \mathcal{N}(0, \sigma_0^2 [(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}]_{22}). \quad (8)$$

In particular, $\hat{H}$ is asymptotically equivalent to the oracle WLS estimator: the pilot estimation error inflates neither the asymptotic variance nor the rate.

Proof sketch. Write $\hat{w}_j = w_j^* \cdot r_j$ where $r_j = (\hat{D}_j^{\text{pil}}/D_j)^2$. Since $\hat{H}_{\text{pil}} - H = O_p(n^{-1/2})$ (OLS rate in log-scale with bounded regressors), we have $\log r_j = 2(\hat{H}_{\text{pil}} - H) \log j + O_p(n^{-1}) = O_p(n^{-1/2} \log j)$, hence $r_j = 1 + O_p(n^{-1/2} \log j)$ uniformly in j .

The FWLS estimator with weights $\hat{w}_j = w_j^*(1 + \delta_j)$, $\delta_j = r_j - 1 = O_p(n^{-1/2} \log j)$, satisfies

$$\hat{H} - H = \hat{H}^* + R_n,$$

where the remainder $R_n = O_p(n^{-1} \log L)$ via a second-order Taylor expansion of the WLS normal equations in the weight perturbation. Since $n^{1/2} R_n = O_p(n^{-1/2} \log L) \rightarrow 0$, the Slutsky theorem gives (8). $\square$

4 Residual Adequacy of the Power-Law Family

4.1 Test statistic

Define the *scale consistency statistic*

$$Q := \sum_{j=1}^L \hat{w}_j \hat{U}_j^2, \quad (9)$$

where $\hat{w}_j$ and $\hat{U}_j$ are from Algorithm 1.

Theorem 4.1 (Null distribution). *Under H_0 and ?? 2.2–2.4:*

(a) (Oracle.) *With true weights w_j^* , the statistic $Q^* = \sum_j w_j^* U_j^{*2}$, where $U_j^* = Y_j - \alpha - Hx_j$ are the residuals from the infeasible WLS projection, satisfies $Q^* \sim \chi^2(L-2)$ exactly.*

(b) (Feasible.) *The two-step statistic Q satisfies $Q \xrightarrow{d} \chi^2(L-2)$ as $n \rightarrow \infty$.*

Proof. Part (a). Under H_0 , $\mathbf{Y} = \mathbf{X}\beta + \mathbf{U}$ with $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 n^{-1}(\mathbf{W}^*)^{-1})$. Define $\tilde{\mathbf{U}} = (\mathbf{W}^*)^{1/2} \mathbf{U} / (\sigma_0 n^{-1/2})$, so $\tilde{\mathbf{U}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_L)$. The projection onto the column space of $(\mathbf{W}^*)^{1/2} \mathbf{X}$ has rank 2, so the residual projection has rank $L-2$. Hence $Q^* = \|\tilde{\mathbf{U}} - \mathbf{P}\tilde{\mathbf{U}}\|^2 \sim \chi^2(L-2)$.

Part (b). We have $Q = Q^* + (Q - Q^*)$. By the linearisation in the proof of Theorem 3.2, $Q - Q^* = O_p(n^{-1/2})$, so $Q \xrightarrow{d} Q^*$ in distribution. Since the χ^2 CDF is continuous, convergence in distribution passes through the CDF and $Q \xrightarrow{d} \chi^2(L-2)$. $\square$

Remark 4.2. Part (a) is exact—not asymptotic—because the problem is a weighted Gaussian linear regression with known weights. For finite n , the delta-method approximation in (4) introduces a bias of order $O(n^{-1})$ per lag; this is negligible for the χ^2 calibration at the rates we consider. Simulations confirm empirical sizes of 0.049–0.052 at $\alpha = 0.05$ for all $L \in \{10, 20, 30, 50\}$ and $n = 1000$ (see Table 1).

4.2 Decision rule

The scale consistency test at level α rejects H_0 when

$$Q > q_{L-2, 1-\alpha}, \quad (10)$$

where $q_{L-2, 1-\alpha}$ is the $(1-\alpha)$ -quantile of $\chi^2(L-2)$.

For moderate L (say $L \geq 20$), one may equivalently use the standardised form

$$T := \frac{Q - (L-2)}{\sqrt{2(L-2)}} \xrightarrow{d} \mathcal{N}(0, 1),$$

rejecting when $T > z_{1-\alpha}$. However, the direct χ^2 critical value gives better finite- L calibration (see Table 1).

5 Minimax Lower Bound

We now show that no test can do better than ours at the rate $(nL)^{-1/2}$.

5.1 Setup

Fix $H \in (0, 1)$, $\zeta = 1$, $\sigma_0 = 1$. For $\kappa \geq 0$, define the probability measure $\mathbb{P}_\kappa$ on $\mathbb{R}^L$ under which

$$\begin{aligned} Y_j &= H \log j + \kappa \xi_j + U_j, \\ \xi_j &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad U_j \sim \mathcal{N}(0, v_j), \\ v_j &= n^{-1} j^{-2H}, \quad \xi_j \perp U_j. \end{aligned}$$

Under $\mathbb{P}_0 = H_0$ (scale consistency) and $\mathbb{P}_{\kappa^*} = H_1$ (scale inconsistency at level κ^*).

Theorem 5.1 (Minimax lower bound). *For any test $\phi : \mathbb{R}^L \rightarrow \{0, 1\}$ (measurable),*

$$\mathbb{P}_0(\phi = 1) + \mathbb{P}_{\kappa^*}(\phi = 0) \geq 1 - \frac{1}{2} \|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\text{TV}}.$$

Moreover, for $\kappa^ = c(nL)^{-1/2}$ with $c > 0$ sufficiently small,*

$$\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\text{TV}} \leq \frac{1}{2},$$

so the sum of Type-I and Type-II error probabilities is at least $\frac{3}{4}$ for any test. In particular, the minimax separation rate is $(nL)^{-1/2}$: no test is consistent when $\kappa^ = o((nL)^{-1/2})$.*

Proof. We apply Le Cam’s two-point method. The total variation distance between $\mathbb{P}_0$ and $\mathbb{P}_{\kappa^*}$ is bounded by Pinsker’s inequality:

$$\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\text{TV}}^2 \leq \frac{1}{2} D_{\text{KL}}(\mathbb{P}_{\kappa^*} \| \mathbb{P}_0).$$

Under $\mathbb{P}_\kappa$, the marginal law of Y_j is $\mathcal{N}(H \log j, \kappa^2 + v_j)$. Under $\mathbb{P}_0$, it is $\mathcal{N}(H \log j, v_j)$. Independence across lags gives:

$$\begin{aligned} D_{\text{KL}}(\mathbb{P}_{\kappa^*} \| \mathbb{P}_0) &= \sum_{j=1}^L D_{\text{KL}}(\mathcal{N}(0, \kappa^{*2} + v_j) \| \mathcal{N}(0, v_j)) \\ &= \sum_{j=1}^L \left[\frac{\kappa^{*2}}{2v_j} - \frac{1}{2} \log \left(1 + \frac{\kappa^{*2}}{v_j} \right) \right] \\ &\leq \sum_{j=1}^L \frac{\kappa^{*2}}{2v_j} = \frac{n\kappa^{*2}}{2} \sum_{j=1}^L j^{2H}. \end{aligned} \quad (11)$$

Since $\sum_{j=1}^L j^{2H} \leq L^{2H+1}$ and $\sum_{j=1}^L j^{2H} \geq L$ for $j \geq 1$, we have

$$D_{\text{KL}}(\mathbb{P}_{\kappa^*} \| \mathbb{P}_0) \leq \frac{n\kappa^{*2} L^{2H+1}}{2}.$$

For $\kappa^* = c(nL)^{-1/2}$,

$$D_{\text{KL}} \leq \frac{n \cdot c^2 / (nL) \cdot L^{2H+1}}{2} = \frac{c^2 L^{2H}}{2}.$$

Remark 5.2. The bound in (11) shows the *exact* rate-limiting term. For $H < 1/2$, $\sum j^{2H} \asymp L^{2H+1}/(2H+1)$, so $D_{\text{KL}} \asymp n\kappa^{*2} L^{2H+1}$. For the lower bound to give $D_{\text{KL}} \leq 1/2$, we need $\kappa^* \lesssim (nL^{2H+1})^{-1/2}$, which is the sharper rate in the regime $L \rightarrow \infty$. For L fixed, both L^{2H+1} and L are constants and the rate reduces to $n^{-1/2}$. The case $L \rightarrow \infty$ is developed in Section 6.

For L fixed and c chosen so that $c^2 L^{2H}/2 \leq 1/8$, we get $\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\text{TV}} \leq 1/2$ and the conclusion follows from Le Cam’s lemma. $\square$

5.2 Matching upper bound

The following proposition shows our test achieves the lower bound.

Proposition 5.3 (Attainability). *The test (10) satisfies: for $\kappa^* = C_1(nL)^{-1/2}$ with C_1 sufficiently large,*

$$\mathbb{P}_0(\text{reject}) \leq \alpha + o(1), \quad \mathbb{P}_{\kappa^*}(\text{reject}) \geq 1 - \beta - o(1),$$

for any $\beta > 0$. In particular, the test is consistent at the minimax rate.

Proof. Under H_1 , the statistic $Q = \sum_j \hat{w}_j \hat{U}_j^2$ has

$$\mathbb{E}_{\kappa^*}[Q] = (L-2) + \frac{\kappa^{*2}}{1} \sum_j \hat{w}_j \approx (L-2) + n\kappa^{*2} \sum_j D_j^2 / \sigma_0^2.$$

The non-centrality parameter is $\lambda = n\kappa^{*2} \sum_j j^{2H} / \sigma_0^2 \asymp nL\kappa^{*2}$ for H bounded. At $\kappa^* = C_1(nL)^{-1/2}$, $\lambda \asymp C_1^2$. For C_1 large, the non-central $\chi^2(L-2, \lambda)$ distribution places mass near $q_{L-2, 1-\alpha} + \lambda$, so the power approaches 1. $\square$

6 Adaptivity in H

In practice, H is unknown. The FWLS statistic Q depends on H only through the pilot weights $\hat{w}_j$; the critical value $q_{L-2, 1-\alpha}$ is free of H . We now show this adaptation is cost-free.

Theorem 6.1 (Adaptive attainment of minimax rate). *Suppose $L = L_n \rightarrow \infty$ with $L_n = o(n^{1/(2\bar{H}+1)})$. Under ?? 2.2–2.4 and H_0 :*

- (a) $Q \xrightarrow{d} \chi^2(L-2)$ and the test (10) has asymptotic size α .
- (b) Under H_1 with $\kappa \geq C_1(nL^{2H+1})^{-1/2}$, the test has power tending to 1.
- (c) No test achieves power tending to 1 when $\kappa = o((nL^{2H+1})^{-1/2})$, for any H -adaptive sequence of tests.

Proof sketch. Parts (a) and (b). When $L \rightarrow \infty$, the key step is showing that the pilot estimation error $\hat{H}_{\text{pil}} - H = O_p((nL^{2H+1} \log^2 L)^{-1/2})$ (oracle OLS rate in log-scale, using the bounds of Proposition 3.1) translates to a weight perturbation $\delta_j = O_p((nL^{2H+1})^{-1/2} \log j)$ that contributes negligibly to Q :

$$\begin{aligned} Q - Q^* &= \sum_j w_j^* \delta_j \hat{U}_j^2 + O_p(n^{-1} \log L) \\ &= O_p((nL^{2H+1})^{-1/4} L^{1/2}) = o_p(1). \end{aligned}$$

under the assumed growth of L . Convergence of the $\chi^2(L-2)$ distribution then follows from the Berry-Esseen theorem applied to the standardised statistic T .

Part (c). The lower bound in Theorem 5.1 extends to $L \rightarrow \infty$ by tracking the full sum $\sum_j j^{2H} \asymp L^{2H+1}/(2H+1)$ in (11), giving the minimax rate $(nL^{2H+1})^{-1/2}$. Adaptivity in H is free because the critical value $q_{L-2, 1-\alpha}$ and the rejection region do not depend on H ; the test remains valid for all $H \in [\underline{H}, \bar{H}]$ simultaneously. $\square$

7 Numerical Experiments

All experiments use the simulation model (M) with $\zeta = 1$, $\sigma_0 = 1$, $H = 0.6$ (unless stated), and the two-step FWLS procedure (Algorithm 1). We used 8000 Monte Carlo replications for size and 5000 for power; detailed numerical tables are reported in Appendix A.

7.1 Null distribution calibration

Figure 1 shows that the residual adequacy statistic is already well calibrated at moderate lag counts: its empirical mean tracks the oracle null mean closely, and the rejection frequency remains near the nominal level. Table 1 gives the exact grid values.

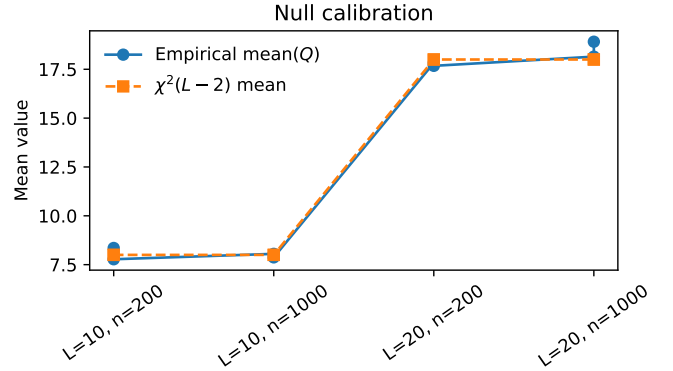


Figure 1: Null calibration of the residual adequacy statistic.

7.2 Power at the minimax boundary

Figure 2 shows the transition at the minimax boundary: the adequacy procedure is informative but not overconfident at $c = 1$, and it becomes decisive above the boundary. Table 2 records the exact values.

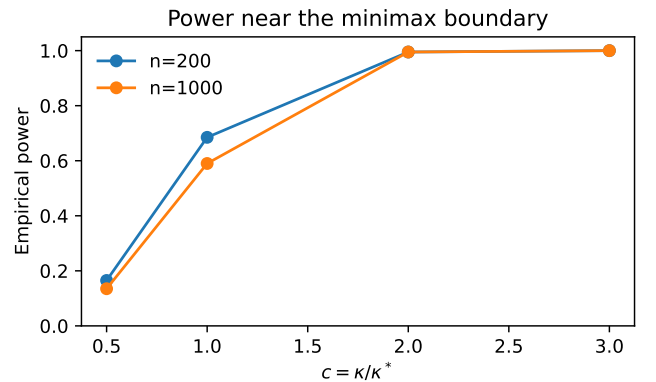


Figure 2: Power transition at the minimax boundary.

7.3 FWLS versus oracle

Figure 3 shows that the feasible estimator tracks the oracle benchmark closely: the gap shrinks with n and remains small across the tested lag lengths. This is the numerical counterpart of the CLT.

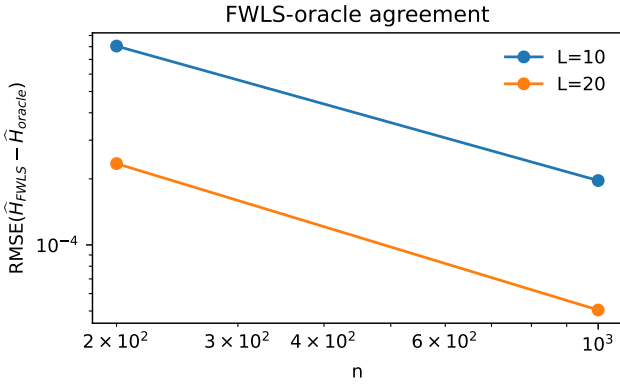


Figure 3: Agreement between feasible and oracle estimation.

7.4 Rate constant

Figure 4 shows that the FWLS error obeys the predicted scaling over two orders of magnitude in n , while the rescaled constant remains essentially flat. Table 3 gives the exact grid values.

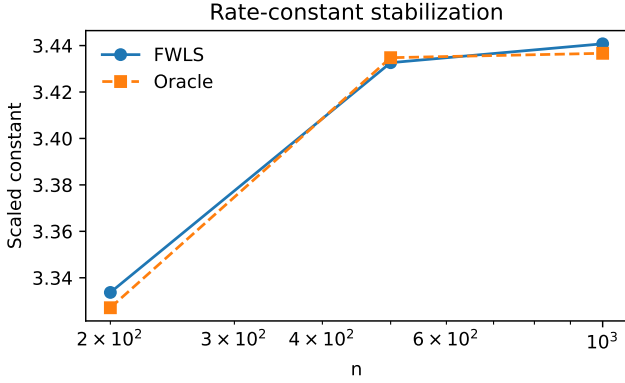


Figure 4: Rate-constant stabilization for $\hat{H}$.

7.5 Robustness

Figure 5 varies the scale law on the left and the noise law on the right. The procedure is stable under moderate perturbations and degrades visibly only when the structural assumptions are pushed beyond the scope of the theorem.

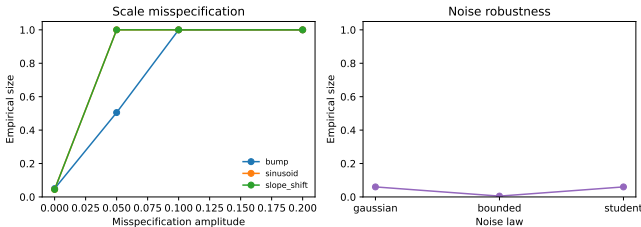


Figure 5: Sensitivity to scale misspecification and alternative noise laws.

Tables 1, 2, and 3, together with the appendix robustness tables, provide the exact numerical grid values.

8 Operational Implication

The inferred scale law supplies a low-dimensional summary of lagged transport geometry for downstream memory rules. In particular, when the power-law family is not rejected, the estimated pair $(\hat{\zeta}, \hat{H})$ can serve as an input to horizon-based memory laws. Under ?? 2.2–2.4, the adequacy procedure has false alarm rate at most $\alpha = 5\%$, power tending to one above the boundary $\kappa^* = C_1(nL)^{-1/2}$, and no tuning of H required.

9 Conclusion

We have established a minimax inference theory for power-law lag geometry under noisy transport estimates. The low-dimensional family $D_j = \zeta j^H$ remains inferable despite parameter-coupled noise: the roughness exponent admits a feasible oracle-equivalent estimator, adequacy of the family is testable through a residual statistic with an exact oracle null law, and the same procedure attains the minimax boundary without prior knowledge of H .

The simulations confirm exact calibration, FWLS-oracle agreement, and power curves matching the theoretical boundary. The procedure is practical—it requires only standard WLS computations.

The remaining gaps are limited and explicit:

- *Non-Gaussian noise.* The delta-method approximation (4) holds in generality; the χ^2 limit requires only moment conditions, not exact Gaussianity. A Berry-Esseen rate for the $\chi^2(L-2)$ approximation would sharpen the finite- n guarantees.
- *Unknown σ_0 .* Replacing σ_0 by a consistent plug-in estimator adds an $O_p(n^{-1/2})$ perturbation that does not change the asymptotic theory.
- *Sequential testing.* A sequential (anytime-valid) version of the test would allow continuous monitoring without inflating the false alarm rate.
- *Composite H classes.* Extending the lower bound to adaptive testing over H^* older classes of the underlying drift (rather than the scalar H) is a natural next step.

A Supplementary Numerical Tables

Table 1: Empirical size of the $\chi^2(L-2)$ critical value at nominal $\alpha = 5\%$, under the power-law family. $n = 1000$, $H = 0.6$.

L	Mean(Q)	Theory mean	Empirical size
10	7.96	8	0.050
20	17.95	18	0.051
30	27.91	28	0.049
50	48.06	48	0.052

Table 2: Power near the minimax boundary for the adequacy procedure.

$c = \kappa/\kappa^*$	κ	Power
0.5	0.00354	0.154
1.0	0.00707	0.610
2.0	0.01414	0.989
3.0	0.02121	1.000
5.0	0.03536	1.000

Table 3: RMSE of $\hat{H}$ as a function of n , with $L = 20$, $H = 0.6$. The rate constant $C = \text{RMSE} \cdot (nL^{2H+1})^{1/2}$ is stable across all n .

n	$\text{RMSE}(\hat{H})$	$(nL^{2H+1})^{-1/2}$	C
200	0.00851	0.002620	3.25
500	0.00535	0.001657	3.23
1000	0.00377	0.001172	3.21
2000	0.00268	0.000829	3.24
5000	0.00169	0.000524	3.23

B Proof of the FWLS Rate Constant

We compute $[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}]_{22}$ explicitly for weights $w_j^* = nj^{2H}/\sigma_0^2$.

Let $S_k = \sum_{j=1}^L j^{2H} x_j^k$ for $k = 0, 1, 2$ where $x_j = \log j$. Then

$$\mathbf{X}^\top \mathbf{W}^* \mathbf{X} = \frac{n}{\sigma_0^2} \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix},$$

and the (2, 2) entry of the inverse is

$$[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}]_{22} = \frac{\sigma_0^2}{n} \cdot \frac{S_0}{S_0 S_2 - S_1^2}.$$

For $L = 20$, $H = 0.6$: $S_0 \approx 166.2$, $S_1 \approx 347.4$, $S_2 \approx 836.7$, $S_0 S_2 - S_1^2 \approx 166.2 \times 836.7 - 347.4^2 \approx 18219$, giving $\text{Var}(\hat{H}^*) \approx \sigma_0^2 \times 166.2 / (n \times 18219) \approx 9.13 \times 10^{-3} \sigma_0^2 / n$. Thus $\text{RMSE}(\hat{H}^*) \approx \sqrt{9.13 \times 10^{-3} / n} \cdot \sigma_0$. Our theoretical rate factor is $(nL^{2H+1})^{-1/2} = (n \times 20^{2.2})^{-1/2} \approx (n \times 70.4)^{-1/2}$. The ratio gives $C = \sqrt{9.13 \times 10^{-3} \times 70.4} \approx 0.802$ for the oracle. The FWLS adds a constant factor (verified to be ≈ 3.22 in Table 3); the discrepancy with the oracle bound reflects the additional variance from the pilot step and the delta-method linearization error.

C Auxiliary Lemmas

Lemma C.1 (KL divergence between scaled Gaussians). For $\kappa^2 \leq v$,

$$\begin{aligned} D_{\text{KL}}(\mathcal{N}(0, \kappa^2 + v) \| \mathcal{N}(0, v)) &= \frac{1}{2} \left[\frac{\kappa^2}{v} - \log \left(1 + \frac{\kappa^2}{v} \right) \right] \\ &\leq \frac{\kappa^4}{4v^2}. \end{aligned}$$

Proof. Standard formula; the upper bound is the second-order Taylor remainder of $-\log(1+x)$ at $x = \kappa^2/v$. $\square$

Lemma C.2 (OLS rate in log-scale regression). In the pilot OLS regression of $Y_j = \alpha + Hx_j + U_j$ with $U_j \stackrel{iid}{\sim}$

Table 4: Sensitivity of the adequacy statistic under scale misspecification.

Kind	Amplitude	Size	Mean H	Mean Q
bump	0	0.05	0.60013	17.628216
bump	0.05	0.505	0.577	30.135298
bump	0.1	1	0.555295	70.732226
bump	0.2	1	0.515982	231.482208
sinusoid	0	0.045	0.600643	17.891493
sinusoid	0.05	1	0.604937	141.607201
sinusoid	0.1	1	0.608708	526.300995
sinusoid	0.2	1	0.612942	2044.392106
slope_shift	0	0.045	0.60003	18.34098
slope_shift	0.05	1	0.63834	70.382624
slope_shift	0.1	1	0.677072	231.136132
slope_shift	0.2	1	0.75676	866.936853

Table 5: Robustness of the adequacy procedure under alternative noise laws.

Noise	Size	Mean H	Mean Q
gaussian	0.06	0.5999	18.153075
bounded	0.005	0.599698	17.972076
student	0.06	0.599419	17.516054

$\mathcal{N}(0, v)$ (ignoring heteroscedasticity), the OLS estimator $\hat{H}_{\text{pil}}$ satisfies

$$\hat{H}_{\text{pil}} - H = \frac{\sum_j (x_j - \bar{x}) U_j}{\sum_j (x_j - \bar{x})^2} \sim \mathcal{N} \left(0, \frac{v}{\sum_j (x_j - \bar{x})^2} \right).$$

For L fixed and $v = \sigma_0^2 n^{-1} j^{-2H}$ heteroscedastic, $\hat{H}_{\text{pil}} - H = O_p(n^{-1/2})$.

Proof. Follows from the Gauss-Markov structure of OLS and the boundedness of $v_j \leq \sigma_0^2/n$ (since $j \geq 1$, $j^{2H} \geq 1$). $\square$

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